

Project Name: Skill-It (Group 4)

Project Members: Braeden, Jeremy, Daniel, Madhav, Ryan

1. Gig / Jobs Object — Braeden

Internal Logical Files (ILFs)

Internal Logical Files (ILFs) are database tables owned and maintained by our system.

Job table

Item	Complexity	FP
Job table	Average	10

The Job table is average because it has 10+ fields and relationships to four other tables.

Category table

Item	Complexity	FP
Category table	Simple	7

Category is simple because it has only 2–3 fields with no complex relationships.

Skill table

Item	Complexity	FP
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Skill table	Simple	7
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Skill is simple because it has only 2–3 fields with no complex relationships.

User Profile table

	Item	Complexity	FP
	User Profile table	Average	10

User Profile is average due to its 12+ fields and references to Skills and MessageThreads.

Review table

	Item	Complexity	FP
	Review table	Average	10

Review is average because it references two UserProfile objects (From and To) plus an associated Job.

External Interface Files (EIFs)

External Interface Files (EIFs) are data sources our system reads but does not own.

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Supabase Auth User

Item	Complexity	FP
Supabase Auth User	Simple	5

The Supabase Auth User table is owned and managed entirely by Supabase's authentication infrastructure. We read it only to verify user identity and never write to it directly, making it an EIF rather than an ILF.

External Inputs (EIs)

External Inputs (EIs) are operations that enter the system and modify data.

Create job posting

Item	Complexity	FP
Create job posting (POST /api/jobs/)	Average	4

Creating a job is average because it writes to the Job table while also referencing Category and Skill.

Edit job

Item	Complexity	FP
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Edit job (PUT /api/jobs/{jobId})	Simple	3
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Simple because it touches only one table with a single, straightforward update.

Delete job

	Item	Complexity	FP
	Delete job (DELETE /api/jobs/{jobId})	Simple	3

Simple because it touches only one table with a single, straightforward delete.

Mark job complete

	Item	Complexity	FP
	Mark job complete (PATCH /api/jobs/{jobId}/complete)	Simple	3

Simple because it touches only one table with a single, straightforward update.

Apply to job

	Item	Complexity	FP
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Apply to job (POST /api/jobs/{jobId}/request)	Simple	3
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Simple because it touches only one table with a single, straightforward update.

Withdraw application

Item	Complexity	FP
Withdraw application (DELETE /api/jobs/{jobId}/request)	Simple	3

Simple because it touches only one table with a single, straightforward delete.

Accept a worker

Item	Complexity	FP
Accept a worker (POST /api/jobs/{jobId}/accept/{userId})	Average	4

Average because it updates AcceptedWorkers and must validate the user against UserProfile.

Remove accepted worker

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Item	Complexity	FP
Remove accepted worker (DELETE <code>/api/jobs/{jobId}/accept/{userId}</code>)	Simple	3

Simple because it touches only one table with a single, straightforward delete.

External Outputs (EOs)

External Outputs (EOs) are operations that produce computed or derived results. This feature has none — all data returned by the API is retrieved directly from the database with no transformation or calculation applied.

External Inquiries (EQs)

External Inquiries (EQs) are read-only operations that retrieve data without modifying anything.

Filter jobs by category

Item	Complexity	FP
Filter jobs by category	Simple	3

Simple because it applies a single filter condition to one or two tables.

Filter jobs by skills

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Item	Complexity	FP
Filter jobs by skills	Simple	3

Simple because it applies a single filter condition to one or two tables.

Filter open/closed jobs

Item	Complexity	FP
Filter open/closed jobs	Simple	3

Simple because it applies a single filter condition to one or two tables.

Jobs by user

Item	Complexity	FP
Jobs by user (GET /api/jobs/user/{userId})	Simple	3

Simple because it applies a single filter condition to one or two tables.

Pending requests for a job

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Item	Complexity	FP
Pending requests for a job (GET /api/jobs/{jobId}/requests)	Average	4

Average because it includes a poster-only authorization check and joins Job with UserProfile.

Reviews for a job

Item	Complexity	FP
Reviews for a job (GET /api/jobs/{jobId}/reviews)	Average	4

Average because it joins Review with two separate UserProfile references (the reviewer and the reviewed).

Threads for a job

Item	Complexity	FP
Threads for a job (GET /api/jobs/{jobId}/threads)	Simple	3

Simple because it is a straightforward filter on the MessageThread table by Job foreign key.

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Total: 98 Function Points

2. Message Threads / Message Text — Braeden

Internal Logical Files (ILFs)

Internal Logical Files (ILFs) are database tables owned and maintained by our system.

Message Thread table

Item	Complexity	FP
Message Thread table	Average	10

MessageThread is average because it has 7 fields and references Users, Job, and Messages while also tracking archived and name state.

Message table

Item	Complexity	FP
Message table	Simple	7

Message is simple because it has only 4 fields.

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Project Members: Braeden, Jeremy, Daniel, Madhav, Ryan

User Profile table

Item	Complexity	FP
User Profile table	Average	10

User Profile is average for the same reasons as in the Jobs feature — 12+ fields with multiple relationships.

External Interface Files (EIFs)

External Interface Files (EIFs) are data sources our system reads but does not own.

Supabase Auth User

Item	Complexity	FP
Supabase Auth User	Simple	5

Same reasoning as the Jobs feature — we read this only to verify who the logged-in user is, and Supabase owns and manages it entirely.

External Inputs (EIs)

External Inputs (EIs) are operations that enter the system and modify data.

Create thread

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Item	Complexity	FP
Create thread (POST /api/threads/)	Average	4

Average because it creates a `MessageThread` record while simultaneously linking a `Job` and setting initial participants from `UserProfile`.

Archive thread

Item	Complexity	FP
Archive thread (PATCH /api/threads/{threadId}/archive)	Simple	3

Simple because it performs a single, targeted update on one table.

Rename thread

Item	Complexity	FP
Rename thread (PUT /api/threads/{threadId}/name)	Simple	3

Simple because it performs a single, targeted update on one table.

Send message

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Item	Complexity	FP
Send message (POST /api/threads/{threadId}/messages)	Simple	3

Simple because it performs a single, targeted update on one table.

Delete message

Item	Complexity	FP
Delete message (DELETE /api/threads/{threadId}/messages/{messageId})	Simple	3

Simple because it performs a single, targeted delete on one table.

Add participant

Item	Complexity	FP
Add participant (POST /api/threads/{threadId}/users/{userId})	Simple	3

Simple because it performs a single, targeted update on one table.

Remove participant

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Item	Complexity	FP
Remove participant (DELETE /api/threads/{threadId}/users/{userId})	Simple	3

Simple because it performs a single, targeted delete on one table.

Mark messages as read

Item	Complexity	FP
Mark messages as read (triggered on thread open)	Simple	3

Simple because it performs a single, targeted update on one table.

External Outputs (EOs)

External Outputs (EOs) are operations that produce computed or derived results. This feature has none — all data returned is retrieved directly from the database with no transformation or calculation applied.

External Inquiries (EQs)

External Inquiries (EQs) are read-only operations that retrieve data without modifying anything.

All threads for logged-in user

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Item	Complexity	FP
All threads for logged-in user (GET /api/threads/)	Average	4
Average because it joins MessageThread and UserProfile filtered by the current session.		

Specific thread + messages

Item	Complexity	FP
Specific thread + messages (GET /api/threads/{threadId})	Average	4
Average because it joins across three tables: MessageThread, Message, and UserProfile.		

Messages in thread

Item	Complexity	FP
Messages in thread (GET /api/threads/{threadId}/messages)	Simple	3
Simple because it joins only two tables with a straightforward filter.		

Participants in thread

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Item	Complexity	FP
Participants in thread (GET /api/threads/{threadId}/users)	Simple	3

Simple because it joins only two tables with a straightforward filter.

Threads for a job

Item	Complexity	FP
Threads for a job (GET /api/threads/job/{jobId})	Simple	3

Simple — a single foreign key filter.

Unread threads

Item	Complexity	FP
Unread threads (GET /api/threads/unread)	Average	4

Average because it joins MessageThread, Message, and UserProfile while filtering by unread status across all three.

Total: 78 Function Points

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Project Members: Braeden, Jeremy, Daniel, Madhav, Ryan

3. Skills – Daniel

Internal Logical Files (ILFs) are database tables owned and maintained by our system.

Item	Complexity	FP
Skills table	Simple	7

Skills are simple, just a primary key int, name, and explicit tag.

Item	Complexity	FP
Profile-Skill-Holder table	Average	10

The skill-holder tables are average complexity since they both have a primary key composed from two foreign keys.

Item	Complexity	FP
Job-Skill-Holder table	Average	10

The skill-holder tables are average complexity since they both have a primary key composed from two foreign keys.

External Inputs (EIs) are operations that enter the system and modify data.

Item	Complexity	FP
Create new skill (POST /api/skill/{skillName})	Simple	3

Creating a skill is simple because it only refers to the Skill table.

Item	Complexity	FP
Make a skill explicit (PATCH /api/skill/{skillId}/explicit)	Simple	3

Making a skill explicit is simple because it only refers to the Skill table.

Item	Complexity	FP
Add skill to a profile (POST /api/pskillHolder/{userId}/{skillId})	Average	4

Adding a skill to a profile is more complex because it requires checking that the skill is not explicit first.

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Item	Complexity	FP
Delete skill from a profile (DELETE /api/pskillHolder/{userId}/{skillId})	Simple	3

Deleting a skill from a profile is straightforward.

Item	Complexity	FP
Add skill to a job (POST /api/jskillHolder/{jobId}/{skillId})	Average	4

Adding a skill to a job has the same complexity as doing so for a profile.

Item	Complexity	FP
Delete skill from a job (DELETE /api/jskillHolder/{jobId}/{skillId})	Simple	3

Deleting a skill from a job has the same complexity as doing so for a profile.

External Outputs (EOs) are operations that produce computed or derived results.

This feature has none — all data returned by the API is retrieved directly from the database with no transformation or calculation applied.

External Inquiries (EQs) are read-only operations that retrieve data without modifying anything.

Item	Complexity	FP
Get a skill by id (GET /api/skill/{skillId})	Simple	3

Getting a skill is super simple, just using the primary key to return the row.

Item	Complexity	FP
Get all skills from a profile (GET /api/pskillHolder/{userId})	Average	4

Getting all skills from a profile requires returning all rows matching a value in one column, so average.

Item	Complexity	FP
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Get all skills from a job (GET /api/jskillHolder/{jobId})	Average	4
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Getting all skills from a job requires returning all rows matching a value in one column, so average.

Item	Complexity	FP
Get all profiles matching certain skills (GET /api/pskillHolder/{skillId}/profiles)	Complex	6

Getting all profiles matching multiple skills is complex, requiring multiple calls to the database, one for each skill. The function would then group by which profiles match the most skills from the input.

Item	Complexity	FP
Get all jobs matching certain skills (GET /api/jskillHolder/{skillId}/jobs)	Complex	6

Getting all jobs matching multiple skills is complex, requiring multiple calls to the database, one for each skill. The function would then group by which jobs match the most skills from the input.

$$27 \text{ (ILF)} + 20 \text{ (EI)} + 0 \text{ (EO)} + 23 \text{ (EQ)} = 70$$

Total: 70 Function Points

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4. Task Related to Reviews

Creating and Posting Reviews

Task breakdown: form renders on webpage → User fills out form and clicks submit → calls method reviews.createReviews → updates table

Internal Logical Files (ILFs) are database tables owned and maintained by our system.

Item	Complexity	FP
Review table	Simple	7
Profile-Review-Holder table	Average	10
Job-Review-Holder table	Average	10

To create and post reviews, we need to update the review table alongside the associated relational tables.

External Inputs (EIs) are operations that enter the system and modify data.

Item	Complexity	FP
Create a review (POST /api/reviews/)	Simple	3

This method would simply update the corresponding table with the

Total: 27 (ILFs)+3 (EIs)=30 Points

Retrieving and Viewing Reviews Based on Profile

Task breakdown: render webpage of profile → retrieves profile → retrieves review based on profile id → lists the profile and the review

Internal Logical Files (ILFs) are database tables owned and maintained by our system.

Item	Complexity	FP
Review table	Simple	7
Profile-Review-Holder table	Average	10
Profile Table	Average	10

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These are corresponding tables needed to look up a profile, then look up corresponding reviews on that profile.

External Outputs (EOs) are operations that produce computed or derived results.

Item	Complexity	FP
Get a Review for user (GET /api/profile/{uid}/reviews/received)	Simple	3
Get a Profile (GET /api/profile)	Simple	3

The second method would retrieve the profile from the profile table, and then use the relation table to retrieve the review of the review table.

Total: 27(ILFS) + 6(EOs) = 33 Points

5. Categories - Madhav

Creating and Adding Categories

Task Breakdown → creating and adding categories to user profiles and job postings.

Internal Logical Files (ILFs) are database tables owned and maintained by our system.

- Category Table (Simple - 7 FP): A standard table containing category names and explicit tags. It is simple because it has a few fields and straightforward relationships
- Category-Job-Holder Table (Average - 10 FP): A relational table linking categories to specific job postings. Similar to other holder tables in our project, this is average complexity due to managing composite key and foreign key relationships.

External Interface Files (EIFs) are data sources our system reads but does not own.

- Supabase Auth User(Simple - 5FP): Our system reads user identity from Supabase's infrastructure to verify permissions for creating or modifying categories, but we do not maintain this data.

External Inputs (EIs) are operations that enter the system and modify data.

- **Create New Category (POST/api/category)** : Adds a new entry to the Category table. It is simple as it touches only one table

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- **(Simple - 3FP)**
- **Link Category to Job (POST /api/categoryHolder/{jobId}/)** : Creates a relationship between a job and a category. This is average as it requires validating both the jobId and the category existence.
 - **(Average - 4 FP)**
- **Make Category Explicit (PATCH /api/category/{categoryId}/explicit)**: Updates a boolean flag on an existing category.
 - **(Simple - 3 FP)**
- **Change Job Category (PATCH /api/categoryHolder/{jobId}/)** : A straightforward update to an existing relationship in the holder table
 - **(Simple - 3 FP)**

External Outputs (EOs) are operations that produce computed or derived results.

EOs are operations that produce derived or computed data for the user.

- **Category Discovery** : Currently, all data in this module is retrieved directly from the database without mathematical transformation or derivation.
 - **(None - 0 FP)**

External Inquiries (EQs)

EQs are read-only operations that retrieve data without modification.

- **Get Category by ID (GET /api/category/{categoryId})**: A direct primary key lookup.
 - **(Simple - 3 FP)**
- **Get Category for a Job (GET /api/categoryHolder/{jobId}/)**: Retrieves the specific category linked to a unique job ID.
 - **(Simple - 3 FP)**
- **Get All Jobs by Category (GET /api/categoryHolder/{categoryId}/jobs)**: Returns a list of all jobs matching a specific category, requiring a column-based filter.
 - **(Average - 4 FP)**

Total Point Calculation:

ILFs: $7 + 10 = 17$

EIFs: 5

EIs: $3 + 4 + 3 + 3 = 13$

EOs: 0

EQs: $3+3+4 = 10$

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Project Name: Skill-It (Group 4)

Project Members: Braeden, Jeremy, Daniel, Madhav, Ryan

6. Account Creation & Profile - Ryan

Project Name: Skill-It (Group 4)

Project Members: Braeden, Jeremy, Daniel, Madhav, Ryan

Internal Logical Files (ILFs)

Item	Complexity	FP
Profile Table	Average	10
Skill Table	Simple	7

The profile table is classified as average because there are 12+ fields and many references to other tables. It is intended to be the core entity of the application. It is directly related to the Supabase Authentication Architecture.

External Interface Files (EIFs)

Item	Complexity	FP
Supabase Auth User	Simple	5

All user verification and authentication will be done using Supabase's built-in authentication system. We will not need to keep track of passwords or login credentials. Each user in this system will have a direct relationship to a profile in the application.

External Outputs (EOs) - This functionality does not require any external outputs. All accesses will be directly output from the system and require no calculations.

External Inputs (EIs)

Item	Complexity	FP
Account Creation	Simple	3
Login	Complex	6
Logout	Simple	3
Update Password	Complex	6
Create Profile	Simple	3
Edit Profile	Simple	3
Verify/Ban Profile	Simple	3

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Add Skill	Average	4
Remove Skill	Simple	4

Most EIs in this module are simple. EIs classified as simple only have effect in the Profile table and perform simple CRUD operations. Adding a skill is listed as average as it may require updates in both the Profile and Skills tables. In the case of Login and Update Password, these functionalities directly use the Supabase Authentication Architecture and are classified as complex for this reason as well as inferring the importance of security for these functionalities.

External Inquiries (EQs)

Item	Complexity	FP
Callback	Complex	6
Get Profile by User	Simple	3
Get Profile by Skill	Simple	3

Getting a user profile by skill and userId are simple as they are only database accesses requiring no calculations. Callback is classified as complex because it requires a long chain of events ultimately handled by Supabase's authentication system. Callback should require little time implementing on our end however.

Total Function Points - 17 (ILFs) + 5 (EIF) + 35 (EIs) + 12 (EQs) = 69